Numerical-method verification: interpretation report ('fast' run)

Run: `numerics_verification_test_run` at run mode `fast`, 2026-09-10.

Preset: `celltype_pairs_sfaEI_Sc0p2sig0p1_noise0p025_dualStd_3cond_mu8p25`

(n = 500, 4000 state variables, piecewise activation, sigma_u = 0.025, three adaptation regimes). Data: `data/numerics_verification_test/numerics_verification/numerics_verification_data.mat`. Figures: `fig_numerics_solver/Fig_numerics_solver.png`, `fig_numerics_lya_method/Fig_numerics_lya_method.png`.

Code:

- `src/analysis/run_numerics_verification.m`
- `src/figures/fig_numerics_verification.m`
- `src/analysis/SRNNNumericsProbe.m`
- `src/model/integrators/coarsen_noise.m`

1. Questions

1. Is the fixed-step stochastic integrator the paper uses (Roessler SRA1 at 400 Hz) precise enough, and does the largest Lyapunov exponent (LLE) it yields depend on the integrator?
2. Do the two Lyapunov methods in the code base, Benettin's reshooting estimate and the QR full-spectrum method, agree on `SRNNCellTypePairs`? They had only been cross-checked on the older `SRNNModel2` class.

2. Methods

2.1 Why trajectories cannot simply be compared

Two of the three regimes have a positive LLE. Any difference between two integrations, however small, grows as exp(lambda t), so a pointwise comparison of whole trajectories measures chaos, not integrator error. The ode45 reference at tolerance 1e-10 is not exempt: it is a true trajectory of the system only locally.

2.2 Reshooting

Precision is therefore measured the way Benettin measures the LLE. A reference trajectory is computed once. Then, at each of many restart times, the tested integrator is reset exactly to the reference state and integrated for one short segment; the distance from the reference at the segment's end is the discretisation error accumulated over that segment, with no chaotic amplification because every segment starts from the truth. Two segment lengths were recorded: 0.02 s, which is Benettin's renormalisation interval (so this is the error over one Benettin segment), and two steps (the shortest span the fixed-step integrator accepts).

The error is reported per state family (dendritic x, adaptation a, depression b) as the root-mean-square over the family's components, and relative to that family's RMS value over the same window.

2.3 The three references

- **Noise-free (sub-experiment A).** Noise off. Reference: ode45 with RelTol = AbsTol = 1e-10, output on a 1600 Hz grid, full n = 500 network. Tested: SRA1 at 400, 800 and 1600 Hz. With the noise off SRA1's drift stages reduce to Ralston's second-order Runge-Kutta, so the segment error should fall 4x per halving of the step (slope 2 on a log-log plot).
- **With noise (sub-experiment B).** Noise on at the preset's sigma_u = 0.025. ode45 cannot integrate the SDE, so the reference is SRA1 itself at 12800 Hz (8x the finest tested rate). The coarser runs consume the SAME Brownian path, rebuilt by exact aggregation: increments are summed, and the second stochastic integral (the area under the path within a step, which SRA1 uses) is carried with its base correction. Decimating the path would have produced a different path with the wrong variance. The difference between runs on one path is the strong error; SRA1 is strong order 1.5 for additive noise, so a slope of 1.5 is the theoretical floor, but where the drift error dominates the slope stays at 2.
- **Lyapunov exponent (sub-experiment L).** Noise off, 400 Hz, 10 s run with 5 s of accumulation. Benettin's LLE with ode45 at 1e-10 for both the fiducial and the perturbed trajectory, versus Benettin with SRA1 for both. Because Benettin uses one integrator for both trajectories most of the discretisation error is common

to the two and cancels in their difference, so the exponents should agree more closely than the raw trajectory error suggests.

2.4 Benettin vs QR (sub-experiment C)

The QR method integrates an N x N variational system per segment and is not feasible at 4000 states. It was run on a reduced network built from the same preset physics: n = 30, in-degree 6 (the preset's connection density), `F_tracks_network = true` so the theoretical spectral radius matches the full network, noise off, ode45, T = [-8, 4] s with 8 s of warm-up and 4 s of accumulation from t = 0. Both methods ran on the same fiducial trajectory. The largest QR exponent should equal Benettin's LLE.

2.5 A methodological trap that was found and closed

The stimulus is a three-step pattern (off / on / off). Its step edges fall at T/3 and 2T/3 rounded to the nearest sample, and the linear interpolant ramps over one sample, so runs at different sampling rates see slightly different input around each edge. A first version of the analysis let the reshoot window straddle an edge, and one edge-crossing segment dominated the RMS: the error fell 3x from 400 to 800 Hz and then 23x to 1600 Hz, the one rate sharing the reference grid. All reshoot windows are now confined to the stimulus-on middle third, where the input is identical on every grid.

3. Results

3.1 Noise-free precision of SRA1 (figure `Fig_numerics_solver`, rows 2-3)

RMS error per component over one 0.02 s segment, and the fitted slope over the 400/800/1600 Hz ladder:

regime	400 Hz	800 Hz	1600 Hz	slope	state RMS (x)	relative at 400 Hz
no adaptation (chaotic)	4.7e-5	1.15e-5	2.9e-6	2.01	3.5	1.3e-5
single-timescale	3.6e-6	9.0e-7	2.3e-7	2.01	0.62	6e-6
multiple-timescale	2.3e-7	5.8e-8	1.5e-8	2.01	0.33	7e-7

Every regime converges at second order, exactly the theoretical order of SRA1's drift. The piecewise activation's derivative discontinuities did not degrade it. Within each regime the x family carries the largest error and b the smallest, which follows from their time constants (x is the fastest state).

3.2 With noise (row 3, dashed)

regime	400 Hz	800 Hz	1600 Hz	slope
no adaptation	5.8e-5	1.4e-5	3.6e-6	2.01
single-timescale	1.3e-5	3.3e-6	8.4e-7	1.98
multiple-timescale	4.6e-6	1.25e-6	3.5e-7	1.84

The noisy error is the same order as the noise-free error and still falls with slope close to 2. At sigma_u = 0.025 the drift error dominates; the strong-order-1.5 stochastic term is not yet visible. (It would appear at much larger noise, and is verified separately in `test_sde_integrators`, which measures 1.73 at large sigma.) The slight droop to 1.84 in the most-adapted regime is the stochastic term beginning to show where the drift error is smallest.

3.3 LLE agreement (row 1 and column titles)

regime	Benettin, ode45 1e-10	Benettin, SRA1 400 Hz
no adaptation	+3.645	+3.715
single-timescale	+0.319	+0.489
multiple-timescale	-0.212	-0.201

In the chaotic and the stable regimes the two integrators agree to 2% and 5%. The single-timescale regime differs by 0.17 on a 5 s accumulation window; see section 4.

3.4 Benettin vs QR on the reduced network (figure `Fig_numerics_lya_method`)

regime	states	Benettin LLE	QR lambda_1
no adaptation	30	-9.898	-9.898
single-timescale	120	-1.100	-0.500
multiple-timescale	240	-0.133	-0.127

4. Interpretation

What the results establish about the integrator.

- At 400 Hz, SRA1 accumulates a relative error of 1e-5 to 1e-7 per state over one Benettin segment, depending on regime, and this error converges at second order, so it can be extrapolated to any step size.
- In the contracting (adapted) regimes the error does not accumulate: it saturates at roughly the per-segment error times the number of segments in one relaxation time. With an LLE near -0.2 that is about 250 segments, giving a pointwise error of order 6e-5 per component after the transient. An SRA1 trajectory at 400 Hz tracks the true one to about 1e-4 there.
- In the chaotic regime a global trajectory error cannot exist for any integrator. Starting from 1.3e-5 relative and growing at 3.6 per second, the error reaches order one after about 3 s, which is where the ode45 and SRA1 traces in row 1 visibly part. A 1e-10 tolerance buys only about three more seconds. Precision claims in that regime must rest on statistics, and the statistic the paper uses is the LLE, which agrees between integrators to 2%.
- One honest caveat: summed over 500 dendritic states the absolute error norm per segment in the chaotic regime is about 1e-3, which is the size of Benettin's perturbation. Benettin is unaffected because both of its trajectories use the same integrator and the error is common-mode; the 2% LLE agreement confirms this empirically.

The single-timescale LLE gap (0.32 vs 0.49). This is a finite-time exponent accumulated over 5 s on a trajectory whose local exponent fluctuates strongly. Two integrators produce two slightly different trajectories through the same attractor and, on a short window, two different finite-time averages. It is not a precision issue in the sense of sections 3.1-3.2 (the reshoot error in this regime is 6e-6). The 'medium' run, which accumulates for 10 s, is the right test.

What the results establish about Benettin vs QR. Less than hoped, for two reasons that both stem from the reduced network.

- The 30-neuron no-adaptation network is a dead fixed point: its LLE of -9.9 is simply -1/tau_d. Agreement there is exact but trivial. Thirty neurons at in-degree 6 cannot sustain activity even with the spectral radius matched, so the reduced network does not reproduce the chaotic regime the paper is about.
- The single-timescale gap (-1.10 vs -0.50) is on a trajectory whose local exponent swings by about +-5 through the whole 4 s window, after the stimulus switches off at t = 0. Both numbers are finite-time values on a transient. Benettin's finite perturbation (d0 = 1e-3) also crosses activation breakpoints that QR's linearisation never sees, so a gap that persists at longer accumulation would be a genuine finding about piecewise nonlinearities rather than a bug.
- The multiple-timescale regime agrees to 0.006.

5. Verdict at 'fast'

- SRA1 at 400 Hz: **verified**. Second-order convergence in every regime, with and without noise; per-segment relative error 1e-5 to 1e-7; LLE integrator-independent to a few percent where the accumulation window is adequate.
- Benettin vs QR on `SRNNCellTypePairs`: **not yet settled**. Exact on a fixed point, within 5% on the most-adapted regime, and a factor of two apart on the single-timescale regime at 4 s of accumulation. Requires the 'medium' run (10 s accumulation, 40 neurons) before any statement is made.

6. Reproduction

```
setup_paths();
numerics_verification_test_run                              % ~11 min at 'fast'
% or the pieces:
run_numerics_verification('run_mode', 'fast', 'out_dir', ...)
fig_numerics_verification('run_dir', 'data/numerics_verification_test', 'variant', 'solver')
fig_numerics_verification('run_dir', 'data/numerics_verification_test', 'variant', 'lya_method')
```

Tests: `scripts/tests/test_numerics_probe.m`, `scripts/tests/test_sde_integrators.m`, `scripts/tests/test_run_modes`